MoCHII Atomic-Data Fitting: CHIANTI Cooling and n-level Fits

Abstract

MoCHII evaluates all metal-line atomic data from analytic fits; the raw CHIANTI database is never parsed at runtime. The Python fitting scripts under `tools/fitting/` are the auditable source of every coefficient in `data/atomic/`. This memo documents the fitting procedure and its two products: the cooling-function fits that carry the thermal-loop line cooling in the $n_e \to 0$ limit, currently 28 ions of the eleven-element registry — suppressed to the local electron density at run time by a (T, n_e) table built from the n-level solve — and the n-level data files (level energies, A-values, and Chebyshev fits of the effective collision strengths in Burgess–Tully scaled space) that supply the output-time line emissivities. Both are verified against the CHIANTI v11.0.2 source data and against PyNeb, and the cooling fits are additionally checked channel by channel against an independent photoionization code on a fixed gas state (`docs/MoCHII_cooling_analysis.pdf`). The fitted files reproduce direct CHIANTI evaluation to $\lesssim 0.5\%$ in the classic diagnostic ratios; the remaining differences from PyNeb (up to 6–10% for the density diagnostics near their critical densities) trace to PyNeb's independent choice of collision strengths and A-values, not to fit error.

1. Overview: two products, one design rule

The metal-line design requires that adding an ion be a data operation, not a code operation. The fitting scripts therefore produce, for each registered ion, self-contained coefficient files with provenance headers (CHIANTI version, population model, fit range, fit form, maximum fit error), of two kinds:

- **Cooling fits (thermal balance).** A smooth closed-form cooling function $\Lambda_{\rm ion}(T)$, cheap and Newton-friendly inside the thermal-balance iteration, in the $n_e \to 0$ limit of statistical equilibrium. This is the low-density *carrier*: at run time the thermal loop suppresses it to the local electron density with a (T, n_e) table built from the same n-level solve (`par%cooling_model = local_ne`, the default), so the thermal balance and the diagnostic line list are one physics.
- **n-level data (diagnostics).** Level energies, statistical weights, radiative A-values, and fitted effective collision strengths $\Upsilon(T)$ feeding the generic n-level solver, evaluated only at output time on the converged state. These supply the density-dependent emissivities and the diagnostic line ratios that the cooling fits cannot give.

The fitting scripts are seeded from EXHALE: `chianti_cooling.py` (copied from `EXHALE/cooling_data/`) reads the raw CHIANTI ASCII ion files (`.elvlc`, `.wgfa`, `.scups`) directly, with no ChiantiPy dependency, and applies the Burgess–Tully (1992) descaling of the stored scaled collision strengths. CHIANTI v11.0.2 lives at `~/RT_Codes/CHIANTI/dbase`.

2. The cooling function

For a transition $l \to u$ with energy gap ΔE_{lu}, the Maxwellian electron-impact excitation rate coefficient is

$$q_{lu}(T) \;=\; \frac{8.629 \times 10^{-6}}{g_l \sqrt{T}}\, \Upsilon_{lu}(T)\, e^{-\Delta E_{lu}/kT} \qquad [\mathrm{cm^3\,s^{-1}}], \tag{1}$$

with $\Upsilon_{lu}(T)$ the effective collision strength obtained by descaling the `.scups` data. In the optically thin, low-density limit every collisional excitation is followed by a radiative decay, so the line cooling normalized by $n_e\, n_{\rm ion}$ is

$$\Lambda_{\rm ion}(T) \;=\; \sum_{l\to u} f_l(T)\, q_{lu}(T)\, \Delta E_{lu} \qquad [\mathrm{erg\,cm^3\,s^{-1}}], \tag{2}$$

where $f_l(T)$ is the fractional population of the lower level. The fitting procedure evaluates the $n_e \to 0$ *limit of statistical equilibrium*: at $n_e \ll n_{\rm crit}$ of the ground-term fine structure the metastable levels are drained radiatively faster than collisions populate them, so every ion sits in its lowest level ($f_1 = 1$) and excitation happens out of that level only. Each excitation is followed by a radiative cascade that ultimately radiates the full ΔE_{1u} whatever route it takes down, so $\Lambda_{\rm ion}$ is the excitation power out of the ground level and is exactly what the n-level solve of Section 4. returns as $n_e \to 0$. ΔE_{lu} is the observed `.elvlc` level difference in both the Boltzmann factor of Eq. (1) and the radiated energy of Eq. (2); the `.scups` de is the Burgess–Tully *scaling* energy and enters only the scaled temperature. Detailed balance ties the excitation rate to the same level gap that appears in the equilibrium matrix, so the two energies must not be interchanged.

3. Cooling-function fits

Each $\Lambda_{\rm ion}(T)$ curve is fitted over 10^3–10^5 K with the multi-exponential form used throughout EXHALE,

$$\Lambda_{\rm ion}(T) \;=\; T^{-1/2} \sum_{i=1}^{N} A_i\, e^{-T_i/T}, \tag{3}$$

a Levenberg–Marquardt fit on $\log\Lambda$. In the $n_e \to 0$ limit Eq. (3) is the exact form of Eq. (2) apart from the slow temperature dependence of Υ: the T_i are the excitation temperatures $\Delta E_{1u}/k$ and the A_i their excitation-weighted strengths. The starting ladder is therefore built from the level energies themselves — the transitions out of the ground level are clustered in $\log(\Delta E_{1u}/k)$, weighted by the share of the cooling each carries across the fit range — and the fitter sweeps $N = 2 \ldots 8$ with perturbed restarts, stopping at the first N that holds the maximum error below 1%. Ions with only a few excitation blocks (the Cl ions have four transitions out of the ground term) still need extra terms, because a single exponential has a fixed amplitude and the temperature dependence of Υ has to be carried by superposing neighboring terms. Table 1 lists the fit quality; Figure 1 overlays the fits on the CHIANTI curves. The coefficients are written to `data/atomic/cooling_<ion>.txt`.

A caveat carried in every file header: Eq. (3) is low-density-limit cooling. Transitions with low critical density (the fine-structure IR lines, most sharply [C II] 158 μm with $n_{\rm crit} \sim 50$ cm^{-3} and the [O III] 52, 88 μm pair; [O II] $\lambda\lambda$3726, 3729; [S II] $\lambda\lambda$6717, 6731) saturate above $n_{\rm crit}$, where the fit overestimates the cooling. The thermal loop therefore treats Eq. (3) as the $n_e \to 0$ carrier and suppresses it to the local density: on the Cloudy c25.00 state of the HII40 benchmark ($n_e \simeq 110$ cm^{-3}) the fitted line cooling exceeds the Cloudy inventory by 11.6%, of which +5.1 points come from [O III] and +3.9 from [C II], while the same ions evaluated with the n-level solve at the local n_e land within 2.4% and 6% and bring the summed line cooling to 1.01 times Cloudy's. MoCHII therefore builds a (T, n_e) suppression table from the same n-level solve at setup and applies it in the thermal loop by default (`par%cooling_model = local_ne`); `low_density` uses the fits directly. The emissivities themselves always come from the n-level data.

Table 1: Cooling fits, 10^3–10^5 K. The maximum relative error is against the direct CHIANTI evaluation of Eq. (2).

Ion	N	max error	output file
O II	4	0.27%	`cooling_o_2.txt`
O III	5	0.32%	`cooling_o_3.txt`
N II	5	0.66%	`cooling_n_2.txt`
S II	3	2.23%	`cooling_s_2.txt`
S III	5	0.23%	`cooling_s_3.txt`
S IV	3	0.61%	`cooling_s_4.txt`
Ne II	5	0.40%	`cooling_ne_2.txt`
Ne III	5	0.34%	`cooling_ne_3.txt`

4. n-level data with fitted collision strengths

For each ion the lowest n levels are exported — enough for the optical/IR diagnostics (Table 2); extra levels are harmless to the solver. The file `data/atomic/nlevel_<ion>.txt` contains three blocks:

- `NLEV`: level index, statistical weight g, energy in cm^{-1} (observed values preferred, theoretical as fallback);
- `NRAD`: radiative rates A_{ul} from `.wgfa` (duplicate rows summed);
- `NUPS`: one line for each collisional transition — the Burgess–Tully type, scaling parameter C, transition energy in Ryd, the fitted domain $[s_{\rm lo}, s_{\rm hi}]$, a log flag, and the Chebyshev coefficients, followed by the recorded maximum fit error.

4.1 Burgess–Tully scaling and Chebyshev fits

CHIANTI stores Υ on the Burgess–Tully (1992) scaled temperature $s \in [0,1]$. With $e_t \equiv kT/\Delta E$, the forward map is

$$s \;=\; \begin{cases} 1 - \dfrac{\ln C}{\ln(e_t + C)}, & \text{types 1, 4,} \\[2ex] \dfrac{e_t}{e_t + C}, & \text{types 2, 3, 5, 6,} \end{cases} \tag{4}$$

and the descaling back to Υ is

$$\Upsilon \;=\; y\ln(e_t + e)\;[1]; \quad y\;[2]; \quad \frac{y}{e_t+1}\;[3]; \quad y\ln(e_t + C)\;[4]; \quad \frac{y}{e_t}\;[5]; \quad 10^{y}\;[6], \tag{5}$$

where $y(s)$ is the scaled value. Instead of the spline-through-table evaluation used by CHIANTI itself, the fitting scripts fit $y(s)$ with a Chebyshev series over the image of $T \in [10^3, 10^5]$ K intersected with the tabulated range,

$$u = \frac{2(s - s_{\rm lo})}{s_{\rm hi} - s_{\rm lo}} - 1, \qquad y = \sum_{k=0}^{n_c - 1} c_k\, T_k(u), \tag{6}$$

clipping s into $[s_{\rm lo}, s_{\rm hi}]$ outside the fit domain. The degree grows from 2 until the descaled $\Upsilon(T)$ matches the direct spline evaluation to 0.2% (cap at 12). Weak intercombination transitions whose tables span decades (e.g. O III $^1D_2, ^1S_0 \to {}^5S_2$)

defeat a linear-space fit; for those the series stores $\log_{10} y$ instead, marked by the log flag (`logf=1`, apply 10^y before descaling). The Fortran `nlevel_mod` must evaluate exactly this recipe; it is replicated in `verify_nlevel_pyneb.py` as the reference implementation.

Table 2: *n*-level exports. "worst Υ error" is the largest maximum relative error among the fitted transitions of that ion, measured on the descaled $\Upsilon(T)$ against the direct spline evaluation over 10^3–10^5 K.

Ion	levels	*A*-values	Υ fits	worst Υ error
O II	5	9	10	0.28%
O III	6	10	15	1.68%
N II	5	8	10	0.17%
S II	5	9	10	1.25%
S III	6	12	15	0.29%
Ne II	2	1	1	0.16%
Ne III	5	9	10	0.19%

The two entries above 1% have benign origins: the O III outlier is the physically negligible $^1S_0$–$^5S_2$ intercombination transition ($\Upsilon \sim 10^{-4}$), and the S II errors reflect the coarse 10-point `.scups` tables of that ion — the deviation is from the interpolating spline, at the level of the resolution of the source data itself.

4.2 Statistical equilibrium and emissivities

Given (T_e, n_e), the solver builds the rate matrix from the file contents alone: collisional rates from Eq. (1) with de-excitation by detailed balance, $q_{ul} = 8.629 \times 10^{-6}\,\Upsilon/(g_u\sqrt{T})$ and $q_{lu} = (g_u/g_l)\,q_{ul}\,e^{-\Delta E/kT}$, plus the radiative A_{ul}; the level populations follow from $\dot{n} = 0$ with the closure $\sum_i n_i = 1$, and the emissivity of a line is $j_{ul} \propto n_u A_{ul}\,\Delta E_{ul}$.

5. Verification

5.1 Internal consistency: fitted files vs. direct CHIANTI

Running the same statistical-equilibrium solver twice — once reading the fitted `nlevel` files, once evaluating the raw CHIANTI splines and `.wgfa` rates directly — isolates the error contributed by the fits and the file format. Over the grid $T_e \in \{5, 10, 20\} \times 10^3$ K, $n_e \in \{10^2, 10^3, 10^4\}$ cm^{-3}:

Diagnostic ratio	worst $\lvert$fit/raw $-1\rvert$
[O II] $\lambda3726/\lambda3729$	0.05%
[S II] $\lambda6717/\lambda6731$	0.52%

The fitted files are faithful to CHIANTI v11.0.2.

5.2 Cross-check against PyNeb

PyNeb (v1.1.31) carries its own atomic data — for these ions the collision strengths SSB14 (O III), T11 (N II), Kal09 (O II), TZ10 (S II) with A-values FFT04-SZ00, FFT04, Z82-WFD96, RGJ19 respectively — so this comparison is a cross-calibration against an independent reference, not a fit-residual test. Tables 3 and 4 give the classic temperature and density diagnostics.

The temperature diagnostics agree to 1.2% ([O III]) and 0.3% ([N II]). The density diagnostics drift to 6–10% at $n_e = 10^4$ cm^{-3}, near and above the critical densities of the ^2D levels, where the ratios become A-value-limited — and the two codes take their A-values from different compilations (CHIANTI v11 vs. Z82-WFD96 and RGJ19). Since the fitted files reproduce raw CHIANTI to $\leq 0.5\%$ (previous subsection), this spread measures the state of the atomic data, not the fitting procedure; it should be kept in mind as the floor of attainable agreement when the validation gates compare MoCHII line ratios with PyNeb-based references.

6. Adding an ion

The procedure is a data operation:

1. add one entry (element, stage, level-energy guesses / level count) to the `IONS` dictionary of `fit_cooling.py` and of `fit_nlevel.py`;
2. run both scripts; check the reported maximum fit errors and the overlay figure;
3. extend `verify_nlevel_pyneb.py` with the diagnostic ratios of the new ion where PyNeb has data, and record the comparison.

The suggested order of addition (cooling importance plus diagnostics) continues with C, Ar, and Fe.

Table 3: Temperature diagnostics vs. PyNeb.

		[O III] $\lambda5007/\lambda4363$			[N II] $\lambda6584/\lambda5755$		
T_e [K]	n_e [cm^{-3}]	MoCHII	PyNeb	diff	MoCHII	PyNeb	diff
5000	10^2	4104	4153	−1.2%	829.6	828.6	+0.1%
5000	10^3	4093	4133	−1.0%	818.3	817.2	+0.1%
5000	10^4	3907	3943	−0.9%	624.7	623.9	+0.1%
10000	10^2	152.0	153.4	−0.9%	66.97	66.93	+0.1%
10000	10^3	151.7	152.8	−0.7%	65.75	65.64	+0.2%
10000	10^4	146.0	147.0	−0.6%	52.23	52.12	+0.2%
15000	10^2	50.97	51.30	−0.6%	29.06	29.08	−0.1%
15000	10^3	50.77	51.10	−0.6%	28.53	28.59	−0.2%
15000	10^4	49.08	49.43	−0.7%	23.26	23.32	−0.3%
20000	10^2	29.82	29.78	+0.2%	19.23	19.23	+0.0%
20000	10^3	29.58	29.67	−0.3%	18.92	18.92	−0.0%
20000	10^4	28.68	28.81	−0.5%	15.71	15.71	+0.0%

Table 4: Density diagnostics vs. PyNeb.

		[O II] $\lambda3726/\lambda3729$			[S II] $\lambda6717/\lambda6731$		
T_e [K]	n_e [cm^{-3}]	MoCHII	PyNeb	diff	MoCHII	PyNeb	diff
5000	10^2	0.8045	0.7641	+5.3%	1.298	1.333	−2.6%
5000	10^3	1.436	1.383	+3.9%	0.8223	0.8396	−2.1%
5000	10^4	2.364	2.491	−5.1%	0.5577	0.5052	+10.4%
10000	10^2	0.7697	0.7411	+3.9%	1.309	1.351	−3.1%
10000	10^3	1.260	1.241	+1.5%	0.8810	0.9139	−3.6%
10000	10^4	2.267	2.397	−5.4%	0.5655	0.5203	+8.7%
15000	10^2	0.7566	0.7341	+3.1%	1.320	1.349	−2.1%
15000	10^3	1.180	1.179	+0.2%	0.9185	0.9520	−3.5%
15000	10^4	2.209	2.345	−5.8%	0.5704	0.5294	+7.7%
20000	10^2	0.7502	0.7319	+2.5%	1.343	1.347	−0.3%
20000	10^3	1.136	1.144	−0.7%	0.9543	0.9786	−2.5%
20000	10^4	2.170	2.311	−6.1%	0.5766	0.5373	+7.3%

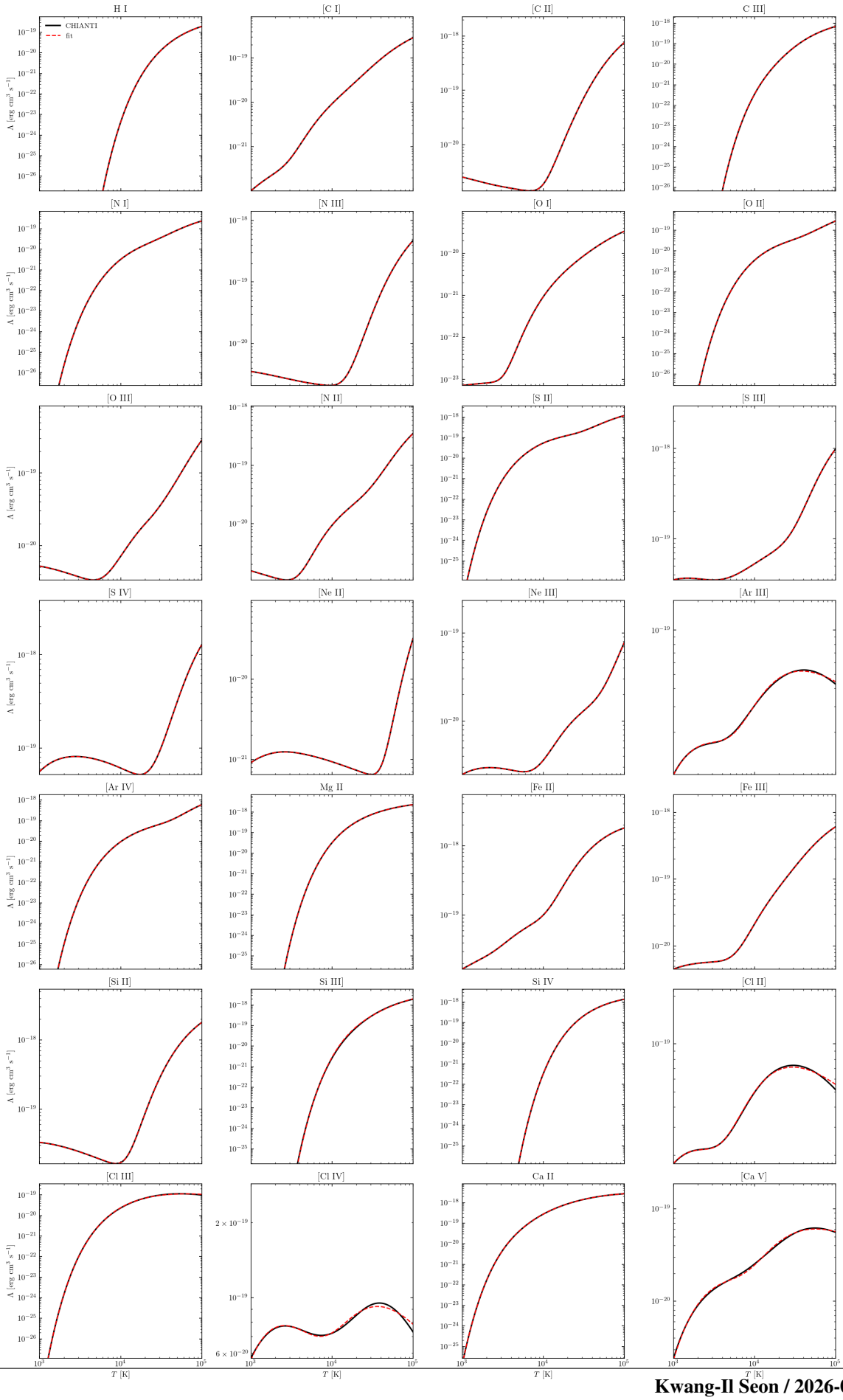


Figure 1: Cooling fits (red dashed) over the direct CHIANTI curves (black), 10^3–10^5 K, in the $n_e \to 0$ limit of statistical